Agent messaging

Accept durably.
Deliver repeatedly until acknowledged.

ELI16 · THE IDEA IN PLAIN LANGUAGE

Registered post: accepting an envelope, showing it to its recipient and receiving a signature are three different events.

Why it matters: A lost reply must not lose accepted work or imply delivery that never happened. Separate mailbox ownership lets execution retry completion safely.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

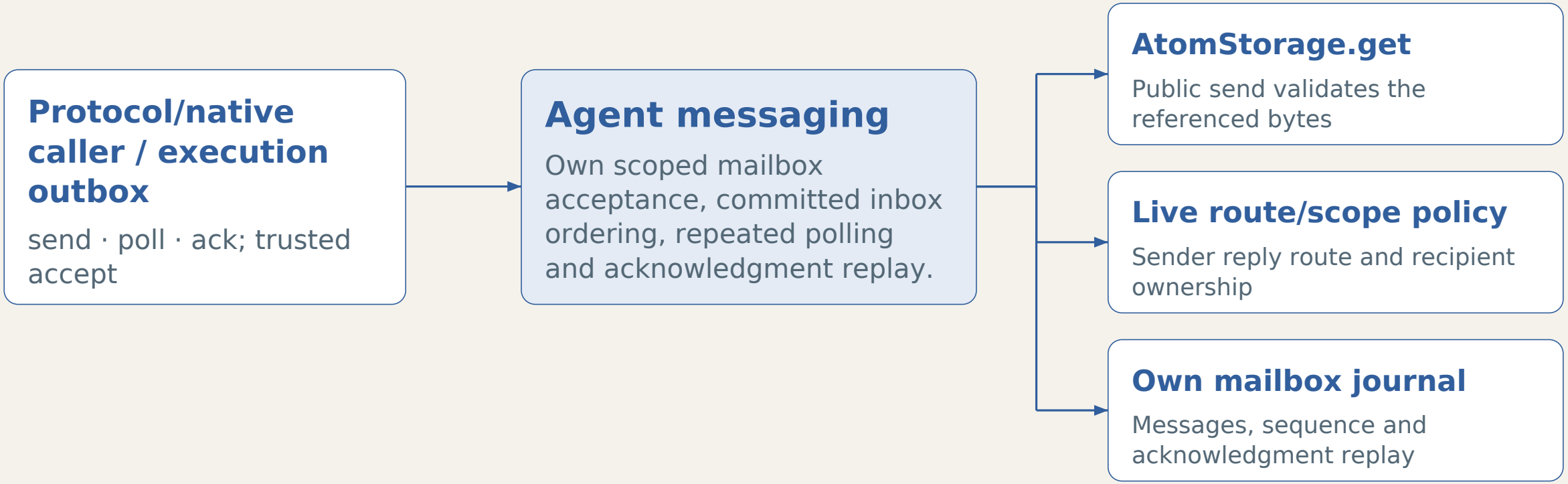
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	**Messaging**
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Own scoped mailbox acceptance, committed inbox ordering, repeated polling and acknowledgment replay.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Inbox + ordering

Durable accepted message, scope/route and committed local sequence

Acknowledgment replay

Recipient acknowledgment identity and stable response

OUTSIDE THIS RESPONSIBILITY

Result payloads and jobs

AtomStorage owns bytes; Execution owns terminal job/outbox state

Exactly-once delivery

Poll may repeat; no global ordering or guaranteed remote arrival

durable-mailbox/2 · explicit legacy profile retained · wire 1.0.0

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
send	Validate reference and registered routes; accept	Acceptance reports delivery not-observed
poll	Return oldest unacknowledged message for owner	Repeats until acknowledged
ack	Commit recipient acknowledgment and replay	Current route ownership still checked
Trusted accept	CompletionDelivery.accept: execution-to-mailbox handoff	Return after mailbox commit; not public dispatch

Finite configured unacknowledged-route capacity; replay records retained for journal lifetime.

A worked example through the boundary

REAL owner persistence + separate worker process · protocol counterparty is a fixture

01 · STEP

Finish job

Execution has a durable completion outbox row.

02 · STEP

Accept mailbox

Mailbox commits that completion independently.

03 · STEP

Lose marking

Worker dies before marking its outbox sent.

04 · STEP

Retry handoff

Same message is accepted once; recipient polls and acks.

What the example establishes

A durable receipt must say which delivery fact was observed.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Acceptance before reply

A success receipt follows the mailbox commit.

I2

Repeat until acknowledgment

Poll observation does not consume the delivery.

I3

Scoped recipient control

Only the registered owner can poll/ack its route.

I4

Semantic replay

Same delivery identity/body replays; changed meaning conflicts.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Mailbox accepted; outbox not marked	Execution retries trusted accept with the same identity.	No duplicate logical message
Acknowledgment reply lost	Retry recovers durable acknowledgment after reauthorization.	No resurrection of consumed delivery
Full / revoked reply route	Outbox remains pending until a valid retry can succeed.	No dropped completion

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

2

Messages in the preserved inspect workflow

One reference share + one completion

0

Pending outbox after recovery demos

Observed logical handoff completion

FIFO

Committed inbox sequence per route

Ordering contract, not global message order

These are delivery/replay observations from shared workflows. No isolated mailbox throughput, delivery p95, multi-consumer scale or network-loss benchmark is recorded.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary backlog, ack delay, replay ratio and route capacity; measure send/poll/ack p50/p95, backlog age and duplicate physical deliveries under fixed semantics.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	SQLite mailbox owner and actual native caller binding; real process restarts.
FIXTURE / LOCAL TEST POLICY	Protocol peer/counterparty, generated messages, local route policy and injected faults.
BLOCKED / UNSUPPORTED	No required current gate blocked; a full or revoked route intentionally delays completion.
UNRUN	Two production agents, remote brokers, distributed ordering, power loss and general messaging federation.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure backlog and replay costs	Rate-controlled sends/acks with fixed capacity	No loss while reporting tail latency and queue age
Qualify a remote transport	Preserve authority, acceptance and acknowledgment meanings	Failure injection across actual network partitions
Specify retention/compaction	Replay lifetime and acknowledgment contract first	No duplicate acceptance after tombstone expiry

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

A durable receipt must say which delivery fact was observed.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m11.py
```

Runs the provider/binding conformance matrix and real DeTTa gate when available. Fixture-only is a distinct mode; it is not native Hive proof.

`fusion/messaging.py`
`fusion/execution.py`
`tests/test_contract.py`
`tests/test_m11.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.